

AttendNet: A Real-Time Deep Learning Framework for Multi-Face Recognition and Automated Attendance Monitoring

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Abstract—Automated face recognition attendance systems offer a contactless alternative to biometric and hardware-based roll-call and biometric attendance systems. Nevertheless, the majority of current solutions are based on single-frame recognition, making them unstable in dynamic scenarios like a classroom where objects are in constant motion, and the lighting is not constant.

In this paper, the author will introduce AttendNet AI, a temporal-sensitive real-time face recognition system that can be used to monitor attendance automatically. The system proposed is an identity matching model that uses a distance-based similarity measure to match identities with a pre-trained deep metric model that generates facial embeddings. To increase the reliability, a multi-frame temporal validation mechanism is added, where the recognition is reliable between consecutive frames before marking attendance.

The system is deployed as a full-stack application with built-in web interface, secure administration authentication, student portal and database based attendance logging. Experimental analysis shows good performance with an accuracy of 96.3, a precision of 96.9, a recall of 95.0 and an F1-score of 95.6.

The findings reveal the enhanced resilience to motion, occlusion, and transient detections, and the system can be deployed in the real world in educational settings.

Keywords - Facial Recognition, VGG Face Model, Real Time Attendance system, Multi- Face Recognition, Temporal validation, Deep learning.

I. INTRODUCTION

In learning institutions, attendance tracking is very important because it shows the participation, engagement, and disciplines of students. The manual attendance systems of roll-call take a lot of time and have the possibility of errors like proxy attendance and wrong record keeping. Even though biometric systems have been developed like fingerprints scanners and

RFID based solutions which are able to automate attendance, they need specific hardware, physical interaction and may not be scalable in large classroom settings.

As AI and computer vision continue to rapidly evolve, face recognition has become one of the potential alternatives to automated attendance systems that is contactless. Deep learning techniques have also proven to be very accurate at facial recognition by deriving feature representations that are discriminative. Most current systems, however, are based on single frame recognition, which is not reliable in real world scenarios where the subject moves a lot, the lighting changes and faces are partially blocked. Such limitations tend to cause false detection and wrong attendance marking.

In order to overcome these issues, this paper presents AttendNet AI, a temporal-aware real-time face recognition system, which combines multi-frame validation with embedding-based recognition. The method suggested will make sure a face is always detected in successive frames before identity is verified and therefore false positives due to temporary detections are greatly minimized. This enhances the strength and reliability of the system in the dynamic classroom environment.

Besides the recognition module, the system is developed as a full-stack application which consists of web-based interface, secure administrator authentication, student portal and database-based attendance control. This integration is a step towards the gap between theoretical face recognition models and real-world implementation.

The main contributions of this work are summarized as follows:

1. A time validation system that ensures multi-frame coherence in trustworthy attendance recording.
2. Real-time multi-face recognition system, which is deep embedding based.
3. An end-to-end application that combines the user interface, database, and analytics to be deployed in real-world scenarios.
4. The validation of high accuracy and strength under experimental conditions in the real-world classroom setup.

II . RELATED WORK

Automated attendance systems have developed a lot as compared to manual and hardware based systems to smart vision-based systems. Older systems like roll-call and biometric systems like fingerprint and RFID-based systems enhance accuracy, but need specialized hardware and require physical interaction which restricts capacity to scale in large settings [1].

The early face recognition methods were appearance-based methods like Eigenfaces and Fisherfaces which employed Principal Component Analysis (PCA) and Linear Discriminant Analysis (LDA) in the extraction of features [2], [3]. All these methods were very sensitive to illumination, pose, and occlusion changes, which limited their usefulness in natural environments, yet, they were computationally efficient.

Scale-Invariant Feature Transform (SIFT) and Histogram of Oriented Gradients (HOG) are feature-based approaches that were proposed to enhance robustness [4], [5]. These methods compute a local invariant features and perform better under moderate variations but, nevertheless, fail in unconstrained settings.

Face recognition was greatly improved with the advent of deep learning. DeepFace showed close to human accuracy with deep neural networks that were trained on large scale data sets [6]. FaceNet also enhanced scalability by embedding faces into a discriminative embedding space, which made identity matching possible with distance measures but did not require retraining [7]. Likewise, VGG-Face employed deep convolutional network to produce high discriminative facial representations [8].

The latest innovations in deep architectures such as ResNet and Inception networks have made recognition more accurate, by allowing deeper and more efficient feature extraction [9], [10]. Moreover, models with low-weight like MobileNet have been introduced to

facilitate the deployment on resource-constrained devices [11].

Simultaneously, object detection models like Fast R-CNN and YOLO have also been used to further improve real-time face detection, making it possible to detect multiple faces in video streams efficiently [12], [13]. Such innovations have made it easier to create real-time facial recognition systems.

A number of recent papers have implemented face recognition to automated attendance systems [14]-[18]. The available methods, however, predominantly use single-frame recognition, which has made them vulnerable to false positives, due to motion, occlusion, and temporary appearance of people in the camera frame. There are approaches which seek to enhance accuracy by using many enrollment images or threshold tuning, but which do not explicitly impose temporal consistency.

Recent articles have investigated video-based recognition and multi-frame analysis to enhance stability [19]-[22]. Although these methods improve performance, they tend to add complexity to the computations or demand state-of-the-art hardware.

On the contrary, the new system proposes a time-sensitive validation system that guarantees the consistency of identity over multiple frames before recording the attendance. The system, in contrast to current techniques, is a lightweight, full-stack solution that can operate in real-time on CPU-based hardware, and is therefore suitable to be practically deployed in education settings.

III . METHODOLOGY

The proposed framework, AttendNet AI, is an AI-based attendance monitoring system, which is implemented using face recognition and is applicable in dynamic classroom settings. The system combines deep embedding based recognition system with a temporal validation system and a full stack deployment system to achieve accurate and reliable attendance marking.

3.1 System Overview

The system works as a pipeline processing system which receives live video input and identifies several faces, facial embeddings, identity matching, and later verifies the recognition of the faces on the sequential frame before recording the attendance.

The proposed method is temporally consistent, unlike traditional methods that use single-frame detectors, which is much more resistant to motion, occlusion, and temporary detections.

3.2 Live Video Acquisition

A live video stream is recorded with a standard RGB camera and separated into successive frames:

$$\{F1, F2, F3, \dots, F_t\}$$

The independent processing of each frame is done with a temporal continuity between the frames.

3.3 Face Detection

A Histogram of Oriented Gradients (HOG)-based detector that is provided by the dlib library is used to detect faces in each frame. The detector allows multi-face recognition and recognizes multiple areas in a frame:

$$D = \{d_1, d_2, \dots, d_n\}$$

where each d_i represents a detected face.

3.4 Face Encoding (Deep Metric Learning)

Every identified face is converted into an embedding of a fixed-length with a pre-trained deep metric learning model:

$$E_i = f(d_i)$$

where:

- d_i is the detected face
- $f(\cdot)$ is the embedding function
- E_i is a 128-dimensional feature vector

The model relies on a deep residual network that is trained to project the faces of the same identity to be closer but different identities are distant in feature space.

3.5 Identity Matching

Identity checks are conducted by comparing extracted embeddings to stored embeddings with Euclidean distance:

$$d(E_a, E_b) = \sqrt{\sum (E_{a,k} - E_{b,k})^2}$$

When the calculated distance is less than a pre-established distance threshold T , the identity is declared a match and otherwise, the face is termed unknown.

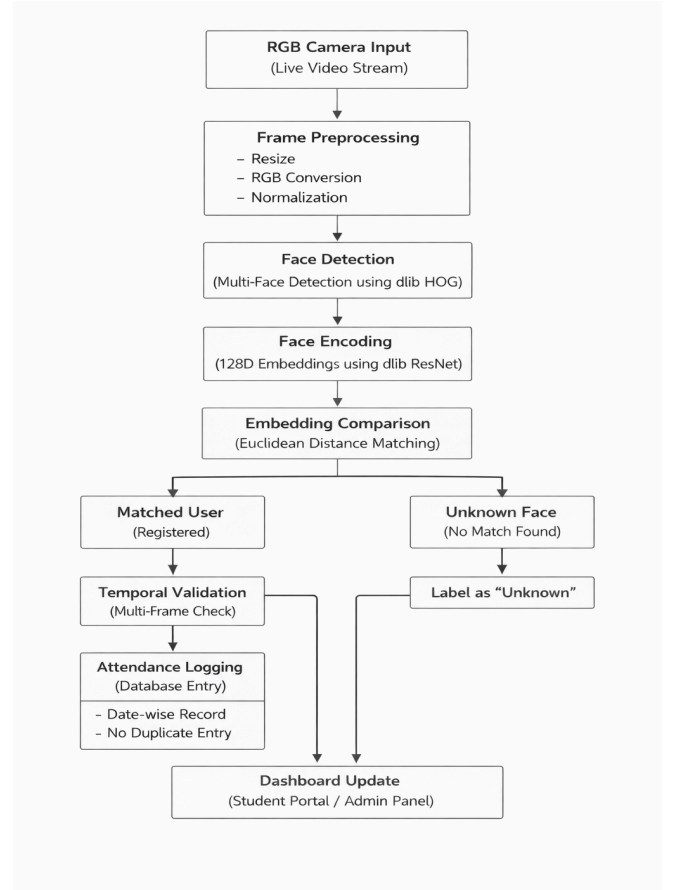


Figure 1. Architecture Diagram

3.6 Temporal Validation Mechanism

A multi-frame temporal validation strategy is proposed to increase the reliability. The system uses recognition consistency between successive frames as opposed to predicting using one frame.

Let C_j denote the number of consecutive frames in which user j is recognized. The attendance is recorded when:

$$C_j \geq N$$

where N is a pre-specified validation threshold.

This system makes sure that only stable detections can be marked as attended to greatly minimize the false positives that might come about through momentary appearances, motion blur, or partial occlusions.

3.7 Attendance Logging and Database Integration

Once the temporal validation condition is satisfied, attendance is recorded in a structured database. The system ensures:

- Unique entry per session
- Automatic date-wise logging

- Prevention of duplicate entries

The ability to integrate with a database allows keeping attendance records in an efficient way, retrieving and managing them.

3.8 System Integration and User Interface

The suggested system is deployed as a full-stack application that comprises:

- Admin authentication module for secure access
- Student registration interface for dataset creation
- Real-time attendance dashboard
- Student portal for viewing attendance records

This integration allows a smooth interaction between the user interface and the processing system in the background.

3.9 Handling Unknown Faces

Any face, which cannot be compared to any stored embeddings in the threshold, is recognized as unknown and not counted in attendance records. This will provide integrity to data and will not allow marking of attendance illegally.

3.10 Summary

The suggested framework integrates deep embedding-based recognition with time validation and complete-stack implementation to give a strong, scalable, and real-time attendance resolution applicable in real classroom settings.

IV . EXPERIMENTAL SETUP

The proposed AttendNet AI system was tested in a real-time classroom-like setting to test the recognition accuracy as well as system behavior in real-life conditions like movement of subjects and change in lighting.

4.1 Implementation Environment

Python was used to implement the system. The processing of real-time video capture and image processing was done using OpenCV and facial embeddings were created using a pre-trained deep metric learning model. The attendance records were kept in a structured database to be easily managed and retrieved.

The experiments were run on a standard personal computer that has 8 GB RAM, and it is not accelerated by a graphics card. A 720p RGB webcam was used to capture video input. No special conditions of lighting or controlled background were employed, so there was an evaluation under the conditions of a real classroom.

The system was capable of recognizing and discriminating a number of faces within a frame. In live testing, a number of respondents were seen in front of the camera at the same time.

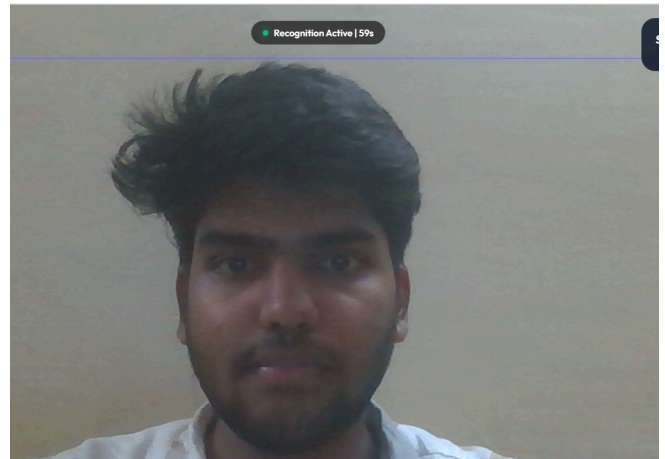


Figure 2 . Real-time face recognition during attendance marking

4.2 Dataset Preparation

A registration phase preceded the live testing where a total of 90 individuals were registered to the system. To guarantee a strong generation of embedding, several face pictures were taken of each participant with different poses and facial expressions.

The same 720p webcam with no controlled conditions was used to take all of the images in natural light. A distinct registration ID was issued to every participant and images were taken and processed to create 128-dimensional facial embeddings, which were stored to be used in real-time comparison and marking attendance.

The numerous samples that each user can use can enhance stability of the recognition by taking intra-class variation like change of pose and facial expression. Because the system relies on a pre-trained model based on embedding, no new training was necessary, and new users can be dynamically added without the need of retraining the model.

Registration
Complete your face profile for AI attendance.

FULL NAME
e.g. John Doe

USN / REGISTER NUMBER
e.g. 23BTRCLO00

COLLEGE EMAIL
hame@college.edu

PHONE NUMBER
+91 98765 43210

Figure 3 . User registration interface for dataset creation

4.3 Testing Procedure

Live testing was conducted after registration with continuous video input. Frames were sequentially processed and the single-user and multi-user case were examined.

An embedding was created and compared on each face that was detected to stored embeddings. Marking of attendance was not done directly upon detection. Rather the system needed to have a consistent identification over a series of successive frames before establishing identity.

Bounding boxes and labels were used to identify recognized users and those who could not be matched were marked as Unknown and were not counted in the attendance list.

4.4 Evaluation Metrics

Standard classification metrics were used to assess the system performance:

- True Positive (TP): Correct identification of registered users
- False Positive (FP): Incorrect identification
- False Negative (FN): Missed detections

- True Negative (TN): Correct rejection of unknown faces

The following metrics were computed:

- Accuracy
- Precision
- Recall
- F1-Score

Moreover, the stability of recognition was examined depending on the multi-frame consistency that was made possible by the temporal validation mechanism.

Metric	Value
Accuracy	96.3%
Precision	96.9%
Recall	95%
F1-Score	95.6%

4.5 Real-Time Multi-Face Evaluation

In order to measure multi-face recognition, several people were standing in front of camera at the same time. The system was able to identify and identify several faces in real time without considerably affecting performance.

A slight delay was realized at the start of the system because of loading the models. The system however, when started met stable and efficient real-time processing.

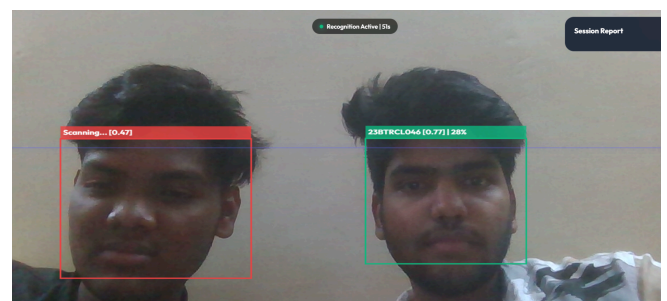


Figure 4 . Attendance records displayed in the system dashboard.

4.6 Attendance Logging Verification

The attendance records generated were analyzed after every session. The system generated entries date-wise and made sure:

- No duplicate entries within the same session
- Accurate marking of present students
- Exclusion of unknown faces

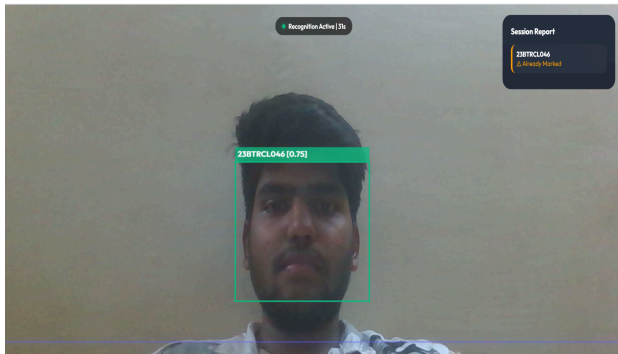


Figure 5 . Attendance records displayed in the system dashboard

4.7 Deployment Observations

Natural movements, variations in posing, and variations in lighting occurred during testing. These differences notwithstanding, the system was able to have a steady recognition owing to the temporal validation mechanism.

The system also proved to be reliable and did not need special hardware or controlled environment to work effectively and thus can be used in the real world.

V . RESULTS AND DISCUSSION

5.1 Overall Performance Evaluation

AttendNet AI system was tested in practice in real-time classroom-like conditions to determine the recognition accuracy as well as the system stability. The system achieved an accuracy of 96.3%, precision of 96.9%, recall of 95.0%, and an F1-score of 95.6%.

High precision means that the system will generate very few false positives and this is important to the attendance systems to avoid false marking. The recall value shows that the majority of registered users are effectively detected, but small differences (e.g. pose changes and partial occlusions) can sometimes occur during detection.

The F1-score is an indicator of a balanced performance of the system in terms of precision and recall.

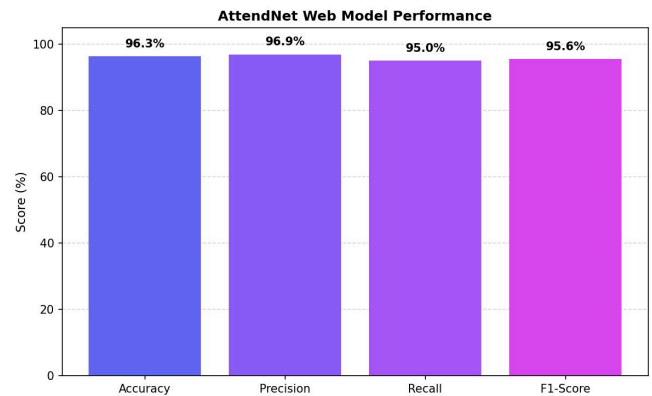


Figure 6 . Performance metrics graph

5.2 Impact of Temporal Validation

The recognition stability was greatly enhanced with the integration of the temporal validation mechanism. In contrast to the traditional systems, which are based on single-frame recognition, the proposed system can guarantee a face is consistently identified (in several consecutive frames) before identity is confirmed.

This minimizes false positive due to temporary detections like temporary appearances or half faces, and improves performance in dynamic conditions when the subject is continuously in motion.

5.3 Real-Time Multi-Face Recognition

The system was experimented in the situations of multiple people at the same time. It was able to identify and match several faces in real time without major impairments on its performance.

Following the initial model loading stage, the system was able to maintain smooth and stable frame processing thus proving its ability to be deployed in real-time.

5.4 System Efficiency

The system is CPU-capable without the use of GPU acceleration. This underscores its appropriateness to be used in resource constrained settings like classrooms.

Though there was a observed slight delay in the initial start up process because of loading of the model, there was a steady performance of the system when it was running continuously.

5.5 Comparison with Existing Models

A comparison was made with the most popular face recognition techniques to determine the effectiveness of the suggested system.

Method	Accuracy (%)	Remarks
Haar Cascade + LBPH	78.3	Sensitive to lighting and pose variations
DeepFace	96.4	High accuracy, requires heavy computation
FaceNet	97.2	Requires large-scale training data
Proposed AttendNet AI	96.3	Real-time, CPU-based, with temporal validation

The findings indicate that more intelligent deep learning networks like FaceNet and DeepFace are more accurate but demand considerable amounts of computing power and data. By comparison, the proposed AttendNet AI system is competitive in accuracy and can run in real-time on standard hardware.

Moreover, recognition stability can also be increased by the introduction of temporal validation which minimizes false positives, which makes the system more practical to implement in practice than the traditional single-frame methods.

5.6 Practical Observations

Participants were found to have natural variations in pose, facial expressions and movement during the time of testing. Regardless of these differences, the system did not exhibit any changes in recognition performance.

The temporal validation mechanism played a crucial role in reducing false positives and ensuring reliable attendance marking. The integrity of the attendance records was preserved because unknown faces were correctly shunned out.

In general, the system exhibits a good balance of accuracy, efficiency, and real-world use, which makes it a viable solution to automated attendance monitoring.

VI. CONCLUSION AND FUTUREWORK

6.1 Conclusion

In this paper, a temporal-conscious real-time face recognition system, AttendNet AI, that is used to monitor attendance in an automated manner was introduced. The suggested system is based on embedding-based recognition and a multi-frame temporal validation system to guarantee reliable identity verification in dynamic settings.

In comparison to the conventional single-frame methods, the system implements recognition consistency on successive frames, which greatly decreases false positives and increases stability. The entire full-stack architecture, which is a web interface, database management, and analytics dashboard, allow a real-world implementation in educational environments.

The experimental scores are quite good, with an accuracy of 96.3, a precision of 96.9, a recall of 95.0 and F1-score of 95.6. It is scalable and available on any hardware with CPU-based resources and does not demand special resources.

In general, the suggested system offers a powerful, effective, and deployable platform of automated attendance-tracking in real-life scenarios.

6.2 Future Work

Despite the great performance of the proposed system, it is possible to make a number of improvements in future work. In order to improve the efficiency and shorten the processing time, the system can be extended with lightweight models of deep learning like MobileNet.

Advanced preprocessing methods and data augmentation approaches can be used to enhance robustness to extreme lighting conditions and to extreme occlusions. Moreover, a centralized database with the cloud-based platform can be integrated to scale the availability of the classes and institutions.

Multimodal authentication, i.e., facial recognition in conjunction with other biometric or contextual data, can also be introduced in future, to enhance reliability and security of the system. Moreover, by optimising

the system to edge devices, deployment can occur on low-power hardware, making it practical in many more situations.

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